

Course One

Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- ☐ Complete the PACE Strategy Document to plan your project while considering your audience members, teammates, key milestones, and overall project goal.
- ☐ Create a project proposal for the data team.

Relevant Interview Questions

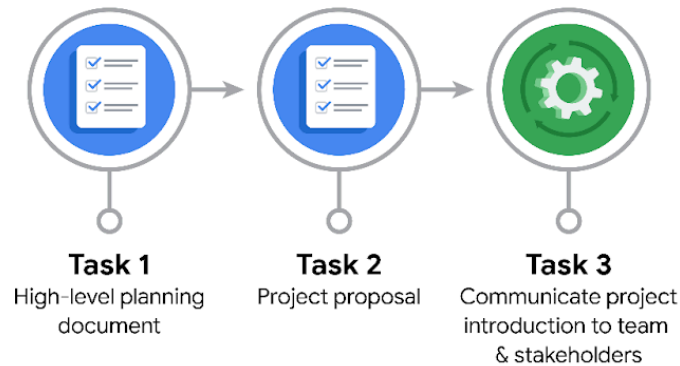
Completing this end-of-course project will empower you to respond to the following interview topics:

- As a new member of a data analytics team, what steps could you take to get 'up to speed' with a current project? What steps would you take? Who would you like to meet with?
- How would you plan an analytics project?
- What steps would you take to translate a business question to an analytical solution?
- Why is actively managing data an important part of a data analytics team's responsibilities?
- What are some considerations you might need to be mindful of when reporting results?



Reference Guide

This project has three tasks; the following visual identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- Who is your audience for this project?

TikTok data team and cross-functional team members.

- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger needs of the client?

I will be working on a predictive model that can be used to determine if a video contains a claim or whether it offers an opinion.

- What questions need to be asked or answered?

What is the condition of the provided dataset?

What variables will be the most useful?

Are there trends within the data that can provide valuable insights?

What steps can I take to reduce the impact of bias?

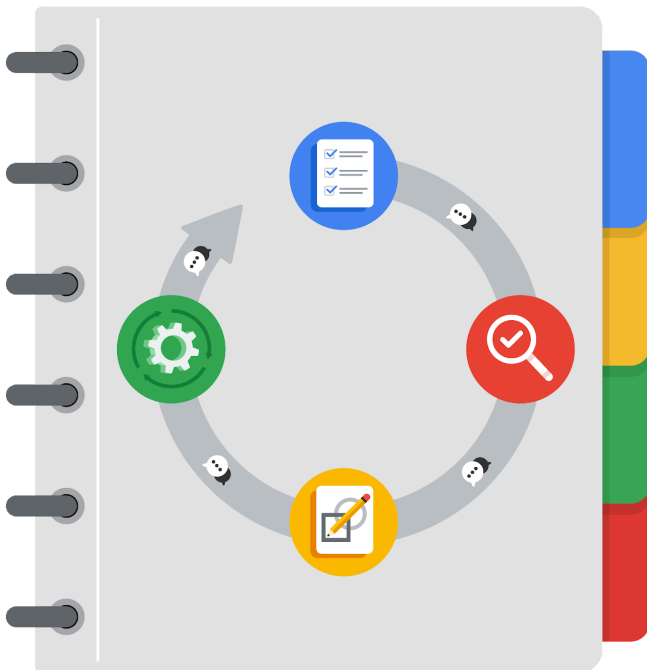
- What resources are required to complete this project?

I will need the project dataset, a python notebook and input from stakeholders.

- What are the deliverables that will need to be created over the course of this project?

I will assist the TikTok team with the development of machine learning techniques that will assist in distinguishing between claims of opinions within context on the TikTok platform. The goal is to build a machine learning model with the capability to reliably classify user interaction data as a claim or opinion.

THE PACE WORKFLOW



[Alt-text: The PACE Workflow with the four stages in a circle: plan, analyze, construct, and execute.]

You have been asked to demonstrate for the company's data team how you would use the PACE workflow to organize and classify tasks for the upcoming project. Select a PACE stage from the dropdown buttons. A few tasks involve more than one stage of the PACE workflow. Additionally, not every workplace scenario will require every task. Refer back to the Course 1 end-of-course portfolio project overview reading if you need more information about the tasks within the project.



Project tasks

Following are a group of tasks your company's data team has determined need to be completed within this project. The data analysis manager has asked you to organize these tasks in preparation for the project proposal document. First, identify which stage of the PACE workflow each task would best fit under using the drop down menu. Next, give an explanation of why you selected the stage for each task. Review the following readings to help guide your selections and explanation: The PACE stages and Communicate objectives with a project proposal. You will later reorder these tasks within a project proposal.

1. **Evaluating the model:** **Execute ▾**

Why did you select this stage for this task?

After the model has been constructed, data is run through to evaluate whether it meets the project's expectation and desired outcomes.

2. **Conduct hypothesis testing:** **Analyze ▾** and **Construct ▾**

Why did you select these stages for this task?

During the analysis phase, it is determined that a statistical test will be used. During the the construct phase, the test is carried out.

3. **Begin exploring the data:** **Analyze ▾**

Why did you select this stage for this task?

During the analysis phase I will gain a deeper understanding of the datasets and the information it contains.

4. **Data exploration and cleaning:** **Plan ▾** and **Analyze ▾**

Why did you select these stages for this task?



During the planning phase the first choices will be made on the methods needed to work with the data. During the analysis phase the data will be cleaned.

5. **Establish structure for project workflow (PACE):** Plan ▾

Why did you select this stage for this task?

Planning stage. Creating an initial project PACE document outlines the work flow and helps to plan how to best approach a project.

6. **Communicate final insights with stakeholders:** Execute ▾

Why did you select this stage for this task?

Communication is necessary at various points throughout the project. Final insights are shared with the stakeholders in the execution phase of the data project.

7. **Compute descriptive statistics:** Analyze ▾

Why did you select this stage for this task?

Evaluating the statistics within data occurs during the analysis phase of the projects,

8. **Visualization building:** Analyze ▾ and Construct ▾

Why did you select these stages for this task?

Visualization begins during the analysis phase and is created during the construction phase.

9. **Write a project proposal:** Plan ▾



Why did you select this stage for this task?

Planning stage. A project proposal is the initial document used to define a project.

10. Build a regression model: **Analyze** and **Construct**

Why did you select this stage for this task?

During the analysing stage, the model is examined in detail to be sure it will meet the needs of the task. The building of the regression morel will take place in the construction phase.

11. Compile summary information about the data: **Analyze**

Why did you select this stage for this task?

Inspecting data to compile information would occur during the analyze phase.

12. Build machine learning model: **Construct**

Why did you select this stage for this task?

The building of the model will take place in the construction phase.